

HFG EVENT

Session 1 : Introduction to electronics= In this session all the basic topic of electronics will be introduced which will be diode, BJT and Mosfet it will be a theory session . In the last 30 min of the event fun activities will be carried out where participants will be given some sets of logic question which should be solved and with each correct answer they can move forward in the snacks and ladders game.

Session 2: Circuit Designing = In this session circuit designing will be thought using the Combination of Transistor and Resistors and also Amplifier Topics will be introduced. In the last 45 min of the event practical implementation will be thought using Tina or LT spice software.

Session 3 : Practical Implementation = In this session participants will implement the real circuit on the breadboard so that they will analyse the circuit more efficiently .

Session 4 : Electronics based fun games = **Circuit Maverick**

The player signs up/logs in and sets up an avatar and profile name. In an office setting, the boss rushes to the player (the protagonist)—the Design Engineer —and gives a list of specifications for a gadget for an esteemed client. The client urgently needs a voltage regulator made with standard components for the launch of a new product (storyline inspired from Stiles (2012)). The circuit is to be delivered urgently with the tightest grip on cost and time. The player must (i) first, design the circuit on a circuit simulator to test the performance (ii) purchase the exact components from a store, (iii) fabricate or made-on-breadboard in the virtual tinkering lab, and then finally (iv) deliver the circuit. The player's final score depends on the time spent at each stage. The game would use game elements such as Experience points (XPs) for overall leaderboard scores and elevating to various levels.